Group representations by recognizable incidence structures

Human Compression Project

Abstract

We recover the finite constructions and free-basis geometries of Leemans, Stokes and Tranchida through the common restriction theorem, an exact shadow-kernel theorem and a free-basis normalizer calculation. The dihedral, symmetric and Platonic examples retain their original attribution and are checked at the exceptional parameters. For ordered projective quadruples, we compute the literal obstruction and give a finite-field replacement whose distinct-length paths retain the ordered endpoints. For the free-group examples, we determine the full type image and the exact natural normalizer. The finite-pair existence theorem of Díaz Ramos, Molinier and Viruel, with its entire proof, is now shared in the common paper. A complete finite certificate and a uniform doubling construction refute the unqualified splitting observation in Section 8.

1 Source and common hypotheses

The source is D. Leemans, K. Stokes and P. Tranchida, *Groups represented by incidence geometries*, arXiv:2506.10464v1, 12 June 2025, thirteen pages. Its incidence systems have a finite set of types, but may have infinitely many objects. Write $C(X)$ for the full correlation group and $K(X)$ for the palette kernel. A representation of a normal pair $H \lhd G$ is a pair isomorphism $(G, H) \cong (C(X), K(X))$. A representation of a group L means the pair $(\mathrm{Aut}(L), \mathrm{Inn}(L))$; a weak representation means compatible injections of these two groups. The geometric condition that every flag extends to a chamber is a separate property. In particular, the abstract's reference to the full group $\mathrm{Out}(L)$ is not the representation definition used in the body: the outer group is the quotient $C(X)/K(X)$ when an exact representation is present.

The restriction-and-extension theorem, the complete shadow-kernel theorem and the unique-shadow corollary of the common paper supply the shared proof. For each finite example we must recognize the old objects under *all* correlations, reconstruct the old incidences, prove pointwise rigidity and exhibit every extension. Counting a chosen group of automorphisms is not by itself a calculation of the full correlation group. For the infinite examples, Theorems R2.Geometry, R2.TypeImage and R2.Normalizer of the common paper provide the complete free-basis argument.

2 The dihedral family

Let $n \geq 3$, let $V = \mathbb{Z}/n\mathbb{Z}$, and put $U_n = (\mathbb{Z}/n\mathbb{Z})^\times$. For $k \in U_n/\{\pm 1\}$ let

$$E_k = \{\{x, x + k\} : x \in V\}.$$

Each E_k is a Hamiltonian cycle. If n is odd, all vertices have one type. If n is even, even and odd vertices have separate types. Each E_k is a further type. Vertex–edge incidence is containment, edges of different types are mutually incident, and in the even case every even vertex is incident to every odd vertex. These are exactly the incidence systems of [L126, §§3.1–3.2].

Theorem R1.Dihedral. For $n > 3$, the full correlation group is the affine group

$$C(\Gamma_n) = V \rtimes U_n, \qquad x \longmapsto ax + b. \tag{D1}$$

Its type-preserving subgroup consists of the affine maps with $a = \pm 1$, with arbitrary b for odd n and even b for even n. This gives the exact pair $(\operatorname{Aut} D_{2n}, \operatorname{Inn} D_{2n})$, where the subscript is the group order. For $n = 3$ it gives the weak representation stated in the source.

Proof. First the vertices are recognizable. For even n, their two types have size $n/2$, while every edge type has size n. For odd $n > 3$, put $d = \varphi(n)/2 \geq 2$. A vertex has degree $2d$, and an edge object has degree $2 + n(d-1)$. Their difference is $(n-2)(d-1) > 0$. Thus every correlation preserves V in both cases. Each edge object is determined by its two endpoints.

The affine maps in (D1) preserve incidence and send E_k to E_{ak}. They act transitively on incident vertex–edge pairs: the image of the pair $(0, \{0, 1\})$ may be prescribed by choosing its vertex image b and one of the two endpoint differences $a \in U_n$. There are $n\varphi(n)$ such pairs. A correlation fixing one pair fixes its edge type, hence acts on its Hamiltonian cycle fixing an edge and both endpoints. An automorphism of a cycle fixing two adjacent vertices is the identity, since successive neighbors are forced. It fixes all vertices and therefore all edge objects. The action on these pairs is thus free as well as transitive, proving (D1).

An affine map fixes every edge type only if it fixes E_1, equivalently $a = \pm 1$; these two multipliers indeed fix every E_k. For even n it fixes both vertex types exactly when b is even. For odd n there is no further constraint.

Write $D_{2n} = \langle r, s : r^n = s^2 = 1,\ srs = r^{-1} \rangle$. Its automorphisms are exactly

$$r \mapsto r^a, \qquad s \mapsto r^b s, \quad a \in U_n, b \in V.$$

Their composition is the affine composition law. Conjugation by r has parameters $(1, 2)$ and conjugation by s has parameters $(-1, 0)$, so the inner subgroup is precisely the subgroup just computed. These formulas also verify compatibility of the two pair isomorphisms.

For $n = 3$ the incidence graph is a hexagon. Its type-preserving group is S_3 and its full group has order twelve. Since $\operatorname{Aut} S_3 = \operatorname{Inn} S_3 = S_3$, the same inclusion into the type-preserving subgroup gives compatible injections. □

Source qualifications. The unit group U_n need not be cyclic. The source's factor $C_{\varphi(n)}$ must be replaced by U_n; for instance $U_8 \cong C_2^2$. The printed equality involving $\operatorname{Out} D_6$ in the small-case proof is also unnecessary: $\operatorname{Out} S_3$ is trivial. Neither correction changes the asserted pair in Theorem 3.1.

For some parameters Γ_n is only an incidence system. At $n = 5$, two differently colored disjoint edges form a flag with no vertex completing it. At $n = 6$, opposite-parity vertices at difference three are incident but have no incident edge object in common. This does not invalidate Theorem 3.1, whose defined notion of representation uses incidence systems. If a genuine finite geometry is required, the pair-preserving cleaning and chamber-completion constructions of Díaz Ramos–Molinier–Viruel [F010] apply. Their complete proofs are included in the common paper; no archived construction theorem is left unpaid.

3 Symmetric groups and the exceptional degree

For $n \geq 2$ let X_n be the rank-two incidence geometry of vertices and edges of the complete graph on n vertices. The point and edge classes have sizes n and $\binom{n}{2}$.

Theorem R1.Symmetric. If $n \geq 4$, then

$$C(X_n) = K(X_n) = S_n.$$

Consequently X_n is an exact representation of S_n for $n \geq 4$, $n \neq 6$. For $n = 3$ it is the preceding weak representation. The literal $n = 2$ instance is not a representation of S_2 in the source's sense.

Proof. For $n \geq 4$ the two type sizes differ. Correlations therefore preserve the vertices. Edges are uniquely recovered from their endpoint sets, and all vertex permutations extend. This proves the displayed pair. We retain the classical theorem $\operatorname{Aut} S_n = \operatorname{Inn} S_n$ for $n \geq 4$, $n \neq 6$, also used in [L126,

Theorem 4.1]. For $n = 2$, both groups of X_2 have order two, whereas both $\operatorname{Inn} S_2$ and $\operatorname{Aut} S_2$ are trivial. Thus the source's parameter exclusion $n \neq 3, 6$ needs a small-parameter qualification. A one-object rank-one geometry supplies the trivial pair when desired. If the parameter $n = 1$ is included, the declared rank-two model has an empty edge type and hence fails the source's surjectivity requirement for the type map; the same one-object rank-one replacement represents the trivial group. □

We can make the exceptional construction exact, without assuming an order calculation for $\operatorname{Aut} S_6$. A *duad* is a two-element subset of a six-element set, and a *syntheme* is a partition into three duads. Form Q with the fifteen duads as points, fifteen synthemes as lines, and containment as incidence, as in [L126, pp.5–6]. Every point lies on a line and every line has three points, so Q is a geometry.

Theorem R1.Six. The duad–syntheme geometry is an exact representation of S_6.

Proof. Distinct duads are collinear exactly when they are disjoint. Hence a type-preserving automorphism preserves disjointness of duads. The largest families of pairwise intersecting duads are precisely the six stars consisting of all five duads through a fixed symbol. Indeed, if an intersecting family has no common symbol, choose $\{a, b\}$ and $\{a, c\}$ in it and then a member avoiding a. That member must be $\{b, c\}$, and any further member must meet all three edges of this triangle, so lies in the triangle. Such a family has at most three members. The six five-member stars therefore recover the six original symbols. Every type-preserving automorphism is the corresponding permutation of symbols; synthemes are then determined by their incident duads. Thus $K(Q) = S_6$.

In the abstract group S_6, the involution classes of transpositions, products of two disjoint transpositions, and products of three have sizes fifteen, forty-five, and fifteen respectively. This follows by choosing the moved symbols and pairing them. Therefore every automorphism of S_6 permutes the first and third classes as a pair. A transposition commutes with a triple transposition exactly when it is one of its three factors: a commuting transposition must exchange two symbols paired with each other by the triple transposition. Consequently the duad–syntheme incidence relation is intrinsically the commuting relation between these two classes. Every automorphism of S_6 acts on Q as a correlation, and this action is faithful because transpositions generate S_6.

Conversely every correlation of Q normalizes $K(Q)$ and hence acts by conjugation on its identified S_6. This conjugation action is faithful. A type-preserving element in its kernel belongs to the trivial center of S_6. A type-swapping element in its kernel would induce an S_6-equivariant bijection between duads and synthemes. Their stabilizers cannot be conjugate: a duad stabilizer contains seven transpositions, whereas a syntheme stabilizer contains only its three constituent transpositions. Thus no such bijection exists.

The two embeddings are inverse because the first action sends each inner automorphism to the natural symbol permutation, and conjugation by its image acts on that subgroup as the original group automorphism. Hence the pair is exactly $(\operatorname{Aut} S_6, \operatorname{Inn} S_6)$. □

The native finite replay also gives $|C(Q)| = 1440$ and $|K(Q)| = 720$. These counts corroborate the proof; they are not needed to deduce the pair isomorphism. We do not rederive the atlas examples for A_6 or A_7 cited but not constructed in [L126]. General finite existence is supplied by common Theorem UR.FinitePairs, the theorem of Díaz Ramos–Molinier–Viruel [F010]. For a finite group L, take $G = \operatorname{Aut}(L)$ and $H = \operatorname{Inn}(L)$, which is normal in G. The A_5 case appears next with a small explicit construction.

4 Cube and octahedron

Regard the cube graph as the Hamming graph on $\{0, 1\}^3$. Give even and odd vertices distinct types, and give edges and square faces two further types. First take the source's literal incidence induced from the cube: vertex–edge, vertex–face and edge–face incidence are containment, and different vertex objects are not incident.

Theorem R1.Cube. The literal rank-four incidence system has pair

$$(C, K) \cong (S_4 \times C_2, S_4), \tag{C1}$$

with the second factor in C generated by the antipodal map. It represents $S_4 \times C_2$ exactly. Adding incidence between opposite-color vertices precisely when they are adjacent cube vertices gives a geometry with the same pair and twenty-four chambers.

Proof. The type sizes are four, four, twelve and six, so every correlation preserves the vertex union, the edges, and the faces. The vertex–edge truncation recovers the cube graph. Edges and faces have distinct vertex shadows, and every graph automorphism preserves their families and the parity bipartition. Recognition therefore identifies the full correlation group with the cube graph group.

Translations of $\{0,1\}^3$ and coordinate permutations give a group of order $8 \cdot 6 = 48$. An automorphism fixing zero permutes its three neighbors. Once these are fixed, each weight-two vertex is the common neighbor of the appropriate pair other than zero, and the weight-three vertex is forced. Thus the displayed group is the full graph group. The parity-preserving subgroup has order twenty-four and acts faithfully on the four even vertices, hence is S_4. Translation by $(1,1,1)$ is central, interchanges parity, and supplies the direct factor in (C1).

For $L = S_4 \times C_2$, its center is the second factor and is characteristic. Every automorphism consequently has the form

$$(s, \epsilon) \longmapsto (asa^{-1}, \epsilon\, \chi(s)),$$

where $a \in S_4$ and $\chi : S_4 \to C_2$ is either trivial or the sign character. The description follows by passing to $L/Z(L) \cong S_4$, using completeness of S_4, and then inspecting the central coordinate. The sign shear is a central involution in $\mathrm{Aut}\, L$, and the other factor is its inner group. Identifying it with the antipodal map proves compatibility of (C1) with the required pair.

The literal system has no chamber, since its two vertex types are never incident. In the repaired system, each chamber consists of an edge, its two endpoints and one of its two incident faces. There are $12 \cdot 2 = 24$. Every flag extends: within a square, two opposite-parity vertices are adjacent; every vertex–edge or edge–face flag extends through an endpoint or a face; and a vertex–face flag extends through either boundary edge at that vertex. These observations also handle singleton and empty flags. The added adjacency is already recovered by the edge objects, so it changes neither group. □

The repair establishes geometry but does not assert firmness: some panels have a unique completion. The source's group-pair theorem survives for its literal incidence system even though its use of the word geometry does not.

5 Dodecahedron, icosahedron and A_5

Use the Petersen graph with vertices the two-element subsets of a five-element set and adjacency given by disjointness. It has ten vertices and fifteen edges. Its twelve pentagons correspond to Hamiltonian cycles of the complete graph on the five symbols: the five edges of such a Hamiltonian cycle are Petersen vertices, and disjointness puts them in pentagon order. Conversely the five pairs in a Petersen pentagon use each symbol twice, since pairs through one symbol form an independent set of size at most two in a pentagon. They consequently form a Hamiltonian cycle on the five symbols.

The natural S_5 acts transitively on these twelve cycles. A cycle stabilizer is a dihedral group of order ten, all of whose permutations on the five symbols are even. Thus A_5 has two orbits $\mathcal{F}$ and $\mathcal{P}$, each of size six, and every odd permutation interchanges them. These are the face and Petrie families in the source's hemidodecahedral construction.

Take vertex, edge, $\mathcal{F}$ and $\mathcal{P}$ as four types. Use containment for their vertex and edge incidences. For a face $F \in \mathcal{F}$ and a Petrie cycle $P \in \mathcal{P}$, set

$$F * P \quad \Longleftrightarrow \quad F \text{ and } P \text{ have a common Petersen edge.} \tag{P1}$$

The rule (P1) is an explicit completion of an incidence left unspecified in [L126, p.7]; it is not quoted as a printed rule.

Theorem R1.Petersen. The completed system is a geometry with 120 chambers and group pair (S_5, A_5). It represents both A_5 and $A_5 \times C_2$ exactly.

Proof. For completeness, the required small cycle calculation has two outcomes:

Pairs in $\mathcal{F} \times \mathcal{P}$	Common vertices	Common edges
30	3	2
6	0	0

Here is the complete six-case calculation with one cycle fixed as $(1, 2, 3, 4, 5)$. We write ab for the Petersen vertex $\{a, b\}$. Rotations and reversal identify Hamiltonian cycles; the six opposite-family cycles and their common Petersen vertices are

12354	12, 23, 45	12435	12, 15, 34
12543	12, 34, 45	13245	15, 23, 45
13524	$\varnothing$	14325	15, 23, 34.

Each listed ordering has odd sign; the twelve Hamiltonian cycles split into two six-element orbits because their dihedral stabilizers consist of even permutations. Thus the list is complete. In each three-vertex intersection, disjointness gives a path of length two. Transitivity on the first family gives the table for all thirty-six pairs. The optional replay enumerates all twelve cycles and verifies the same calculation.

Every Petersen edge belongs to two cycles in each family, by transitivity and the count $6 \cdot 5/15 = 2$. Each incident face–Petrie pair has a common edge, and each of their common vertices lies on a common edge by the table. Hence every flag extends to a vertex–edge–face–Petrie chamber. The number of chambers is

$$15 \cdot 2 \cdot 2 \cdot 2 = 120.$$

The type sizes ten, fifteen, six and six recognize vertices and edges. The graph group is S_5: its largest independent sets are the five four-member stars of pairs through a symbol. As in the duad proof, a pairwise intersecting family with no common symbol has size at most three. Recovering these stars recovers the five symbols. Faces and Petrie objects have distinct vertex supports, so restriction is injective. All of S_5 preserves (P1), while exactly A_5 fixes both cycle families. The recognition lemma gives the asserted group pair.

Finally $\operatorname{Aut} A_5 = S_5$ and $\operatorname{Inn} A_5 = A_5$, a classical retained group fact. For $L = A_5 \times C_2$, the center is characteristic and the quotient is A_5. There is no nontrivial homomorphism $A_5 \to C_2$, since A_5 is perfect. Thus $\operatorname{Aut} L = \operatorname{Aut} A_5 = S_5$, with inner subgroup A_5. The same pair proves both representations. □

Among S_5-invariant completions of the source's two specified truncations, (P1) is the only one that gives a geometry. The thirty and six cross-pairs are the two orbits. To see transitivity on the thirty, fix one Hamiltonian cycle. Its order-five rotation is transitive on the five noncomplementary opposite-family cycles: the only Hamiltonian edge sets fixed by this rotation are that cycle and its complement. Including the six disjoint pairs produces flags having no common point, while excluding the thirty intersecting pairs leaves no chambers. This uniqueness concerns that precise completion problem, not all possible representations of these groups.

6 The literal projective construction

Theorem 6.1 of Leemans–Stokes–Tranchida takes the old objects of $P = \mathrm{PG}(n-1, K)$ and, for each $\lambda \in K \setminus F$, adds the ordered collinear quadruples with cross-ratio λ as a new type Q_λ. An old point is incident with a quadruple exactly when it is one of its four entries; an old higher-dimensional subspace is incident exactly when it contains the four points. Any two new objects of different types are incident. The old shadow therefore remembers the unordered support, not its ordering.

Theorem 1 (Exact obstruction over four elements)**.** *For the literal construction with $n = 3$, $F = \mathbb{F}_2$ and $K = \mathbb{F}_4$, the pointwise-old-object kernel in the type-preserving group is* $\mathrm{Sym}(12)^{210}$*. Hence the literal system is not a representation of* $\mathrm{PGL}(3,4)$*. The same instance refutes the literal root-type truncation of Corollary 6.2.*

Proof. Write $K = \mathbb{F}_2[a]/(a^2 + a + 1)$. The old plane has 21 points and 21 lines, each line containing five points. For any four-subset A of a line ℓ, all 24 orderings have old shadow $A \cup \{\ell\}$. Their cross-ratios are neither zero nor one, so they belong to $\{a, a+1\}$. Interchanging the last two positions inverts the cross-ratio. Since $a^{-1} = a + 1$, this is a bijection between the two sets of orderings. Each type therefore has exactly twelve objects with this shadow.

There are $21\binom{5}{4} = 105$ distinct such shadows, at each of two new types. the complete shadow-kernel theorem of the common paper gives exactly $\mathrm{Sym}(12)^{210}$ as the pointwise kernel. In particular the type-preserving group contains Sym(12), of order 479001600, whereas

$$|\,\mathrm{PGL}(3,4)| = \frac{(4^3-1)(4^3-4)(4^3-4^2)}{4-1} = 60480.$$

Thus even the abstract group orders preclude the proposed isomorphism. There are $42 + 2 \cdot 105 \cdot 12 = 2562$ objects in the literal model.

For Corollary 6.2 choose $\alpha = a$. Its minimal polynomial over $\mathbb{F}_2$ is $X^2 + X + 1$, with precisely $a, a + 1$ as roots in K. For $n = 3$ the displayed truncation keeps both old types and both new types. It is exactly the same counterexample. □

The source proof's uniqueness assertion is the failed step. The always valid double interchange $(p_1, p_2, p_3, p_4) \mapsto (p_2, p_1, p_4, p_3)$ already gives different objects with the same cross-ratio and old shadow. The finite example above supplies the complete kernel and an order contradiction; it does not depend on identifying the full correlation group.

Proposition 2 (Collapsing twins does not restore the field label)**.** *In this four-element example the image of restriction is all $K(P)$. The nontrivial field automorphism has a type-preserving lift to the literal extension.*

Proof. At both new types, every four-subset on every line has the same multiplicity twelve. Every old type-preserving automorphism preserves those multiplicities, so $H = K(P)$ in that theorem. Explicitly, apply Frobenius $x \mapsto x^2$ to all coordinates and all quadruple entries, then interchange the last two entries of each quadruple. The new cross-ratio is $(\lambda^2)^{-1} = \lambda$ for $\lambda \in \{a, a + 1\}$. The two operations commute and each squares to the identity; their composite is an involutory type-preserving lift of Frobenius. □

The ordered-position proposition in the common paper gives a role-preserving ordered-tuple encoding. By itself it does not give a natural projective representation: recognizability under all correlations, extension of duality and the palette action are additional requirements. The distinct-arm construction below discharges them over finite fields. For finite groups L, including each finite projective group, the complete finite normal-pair theorem of Díaz Ramos–Molinier–Viruel, proved in common Theorem UR.FinitePairs, supplies a representation by substituting $(G, H) = (\mathrm{Aut}(L), \mathrm{Inn}(L))$; the inner automorphisms form a normal subgroup of the finite automorphism group. This is that theorem's existence consequence, not a verification of the literal projective model.

Two further scope points are visible in the source definitions. Its preliminary definition requires a finite palette, whereas $K \setminus F$ can be infinite in Section 6. Also the full semilinear group uses all field automorphisms; identifying it with $\mathrm{Gal}(K/F)$ requires those automorphisms to fix F. The finite $\mathbb{F}_4/\mathbb{F}_2$ counterexample satisfies both favorable conditions, so neither qualification removes the obstruction. The orthogonal, degree-two and Suzuki variants in Section 6.2 are proposed investigations, explicitly without the necessary construction details in the source; no natural model for those suggestions is claimed here.

Proposition 3 (A finite-palette obstruction over a rational function field)**.** *For $n \geq 3$, the group $L = \mathrm{PGL}(n, \mathbb{Q}(t))$ has no exact incidence representation with finitely many types, even if infinitely many objects are allowed.*

Proof. For each integer m, substitution $t \mapsto t+m$ is a field automorphism and therefore acts entrywise as an automorphism ψ_m of L. These automorphisms have distinct images modulo $\mathrm{Inn}(L)$. Indeed $\psi_m \psi_k^{-1} = \psi_{m-k}$. If ψ_h were inner, the projective classes of $D = \mathrm{diag}(t, 1, \dots, 1)$ and $\mathrm{diag}(t+h, 1, \dots, 1)$ would be conjugate. Thus one matrix $\mathrm{diag}(t+h, 1, \dots, 1)$ would be conjugate in $\mathrm{GL}(n, \mathbb{Q}(t))$ to a scalar multiple uD. The repeated eigenvalue has multiplicity $n-1 \geq 2$ and the other eigenvalue has multiplicity one, so it forces $u = 1$. The remaining eigenvalue forces $h = 0$. Thus $\mathrm{Out}(L)$ is infinite.

For any incidence system with a finite palette I, the homomorphism from correlations to $\mathrm{Sym}(I)$ has kernel precisely the type-preserving group. Consequently C/K is finite. An exact pair isomorphism with $(\mathrm{Aut}(L), \mathrm{Inn}(L))$ would identify this finite quotient with $\mathrm{Out}(L)$, a contradiction. □

This concerns the definition in the source, whose palette is finite. It is not an obstruction to a different theory allowing infinitely many types, nor does it rule out weak representations.

6.1 A faithful finite-field construction

Here is a positive replacement of the intended finite-field mechanism. It changes the attachment and includes both point and dual pencils, so it is distinct from the literal source construction just refuted.

Let $q = p^f$ be a prime power and $n \geq 3$. Take the full proper nonzero subspace incidence system $P = \mathrm{PG}(n-1, q)$, with dimensions as types. If $f > 1$, choose either

$$R = \mathbb{F}_q \setminus \mathbb{F}_p, \quad \text{or} \quad R = \{\alpha, \alpha^p, \dots, \alpha^{p^{f-1}}\} \quad \text{where } \mathbb{F}_p(\alpha) = \mathbb{F}_q.$$

For $f = 1$ put $R = \varnothing$. For every $\lambda \in R$ take all ordered quadruples of distinct collinear points with cross-ratio λ, and all ordered quadruples of hyperplanes in a pencil with cross-ratio λ. The latter is the usual cross-ratio on the corresponding projective line in the dual vector space. The center color is λ in both cases. Apply the rigid-arms theorem of the common paper with the arm lengths $2, 3, 4, 5$. Write $Y_{n,q,R}$ for the resulting incidence system.

Theorem 4 (Finite projective representation with ordered attachments)**.** *For $n \geq 3$, restriction identifies the pair of the system above with*

$$(C(Y_{n,q,R}), K(Y_{n,q,R})) \cong (\mathrm{P\Gamma L}(n, q) \rtimes \langle \delta \rangle, \mathrm{PGL}(n, q)),$$

where δ is the standard point–hyperplane duality of order two. With the retained classical identification of the full projective-group automorphism group, this is an exact representation of $\mathrm{PGL}(n, q)$*. A finite incidence geometry with the same pair then follows from the complete cleaning and chamber theorems of the common paper.*

Proof. If $f = 1$, no objects are added, and the assertion is the classical fundamental theorem for finite projective spaces, including duality. Suppose $f > 1$, so $q \geq 4$. Every old dimension type has size

$$\begin{bmatrix} n \\ d \end{bmatrix}_q = \prod_{a=0}^{d-1} \frac{q^{n-a} - 1}{q^{d-a} - 1} \equiv 1 \pmod{p}.$$

The denominators are nonzero modulo p, which justifies the displayed congruence of integers. There are $L = \begin{bmatrix} n \\ 2 \end{bmatrix}_q$ point lines and also L hyperplane pencils. For a fixed cross-ratio other than zero and one, a projective line has exactly $(q+1)q(q-1)$ ordered quadruples: choose the first three distinct points, and the cross-ratio determines the fourth uniquely. Thus each center color has

$$M = 2L(q+1)q(q-1)$$

objects; each internal arm color has $M|R|$ objects. Every new type size is divisible by p. Since a correlation permutes entire type classes and preserves their sizes, it preserves precisely the union of old types, characterized by size not divisible by p. The recognition hypothesis of the rigid-arms theorem is proved.

We retain the classical fundamental theorem of projective geometry: the correlations of the full subspace incidence system are the semilinear projectivities and their composites with duality. A semilinear transformation with field part σ sends cross-ratios to $\sigma(\lambda)$. Both choices of R are invariant under every field automorphism. The duality sends an ordered point quadruple to an ordered hyperplane-pencil quadruple with the same cross-ratio, and conversely. This can be seen in coordinates by representing a pencil by its two-dimensional space of linear forms; the standard duality uses the same coefficient vectors. Hence every old correlation preserves the tuple family and permutes its center colors. The rigid-arm theorem supplies its unique extension and rules out further ones.

The internal colors are fixed by every extension. Fixing the old colors rules out duality because $n \geq 3$ and points and hyperplanes are different types. The remaining action has a field part σ; fixing every center color means fixing R pointwise. If R is the orbit of α, this fixes the field generator α, hence $\sigma = 1$. If $R = \mathbb{F}_q \setminus \mathbb{F}_p$, fixing it pointwise also fixes the prime field and therefore all of $\mathbb{F}_q$. Consequently the palette kernel is exactly $\mathrm{PGL}(n, q)$. The converse inclusion follows because projectivities preserve cross-ratios and dimensions.

Finally we retain the standard finite classical-group identification $\mathrm{Aut}(\mathrm{PGL}(n, q)) = \mathrm{P\Gamma L}(n, q) \rtimes \langle\delta\rangle$ and $\mathrm{Inn}(\mathrm{PGL}(n, q)) = \mathrm{PGL}(n, q)$ for $n \geq 3$. These are the group-theoretic inputs identifying the normal pair just proved with the desired pair. The two common conversion theorems apply to the finite properly colored graph $Y_{n,q,R}$ and preserve both groups, giving the geometric version. □

The prime-field generation condition in the smaller label set is intentional. Generating K over a larger prescribed subfield need not force all automorphisms of K fixing the chosen labels to be trivial. The theorem asserts the displayed finite-field construction only. It makes no arbitrary infinite-field representation claim, and it does not identify its new objects with the source's old quadruple objects.

7 The source's conjugate families

First dispatch the small index sets in source Theorem 7.3. For a single freely independent generator, the group is infinite cyclic and its sole omitted-generator subgroup is trivial. Its singleton cosets form a one-type geometry: every chamber is one object and translation acts regularly. There is no residue of rank at least two to check. If the empty index set is allowed, the empty system has the single empty chamber and trivial group. Thus these endpoints satisfy the source flag-transitivity assertion; the following nontrivial family uses the common theory for rank at least two.

Let $F_n = F(x_1, \dots, x_n)$ with $n \geq 2$, and put

$$b_{j,\epsilon,i} = x_j^{\epsilon} x_i x_j^{-\epsilon}, \qquad j \neq i, \quad \epsilon \in \{1, -1\}. \tag{F9}$$

There are $r = 2n(n-1)$ displayed words. Let G be the subgroup they generate.

Lemma R2.Core. The words (F9) freely generate G.

Proof. Construct a finite labeled graph with one central vertex o and $2n$ outer vertices $v_{j,\epsilon}$. Join o to $v_{j,\epsilon}$ by an oriented edge labeled x_j^{ϵ}. At $v_{j,\epsilon}$ put a loop labeled x_i for every $i \neq j$. Map this graph to the rose labeled $x_1, \dots, x_n$.

At each vertex, all outgoing oriented labels are different: at o they are the $2n$ signed letters; at $v_{j,\epsilon}$ the edge back has label $x_j^{-\epsilon}$, and the loop directions have labels $x_i^{\pm 1}$ for $i \neq j$. Thus a reduced edge path maps to a reduced word. A nonempty reduced closed path cannot map to the identity in F_n, so the induced homomorphism on fundamental groups is injective. The spokes form a spanning tree. The associated free basis consists of a spoke, one outer loop, and the reverse spoke, giving exactly (F9).

Equivalently the graph rank is

$$\big(2n + 2n(n-1)\big) - (1 + 2n) + 1 = 2n(n-1).$$

This proves the assertion. □

The finite graph is the core relevant to the displayed finite generating set. Attaching loops at every integer vertex of entire coordinate axes, as a literal reading of the subsequent source description suggests, would instead give infinitely many independent loops. No such enlarged graph is used here. Trees may be attached to the finite core to obtain the subgroup covering without altering its fundamental group.

Theorem R2.SourceFamily. Form the source coset system on G with one maximal standard parabolic omitting each word (F9). Then it is flag-transitive and residually connected, G acts faithfully and regularly on chambers, and its full type quotient is

$$\mathrm{Sym}(2n(n-1)). \tag{F10}$$

The signed permutation group $C_2^n \rtimes S_n$ of the ambient basis acts faithfully by correlations. Its action is a specified subgroup, not the complete type image.

Proof. Lemma R2.Core identifies G with the free group on the displayed basis. Theorem R2.Geometry and Theorem R2.TypeImage apply. An ambient signed permutation $x_k \mapsto x_{p(k)}^{\delta_k}$ sends

$$b_{j,\epsilon,i} \longmapsto b_{p(j),\epsilon\delta_j,p(i)}^{\delta_i}. \tag{F11}$$

It therefore permutes the omitted-generator parabolics and acts by (F6). Its palette action is faithful: fixing all indices (j, ϵ, i) forces p to fix every symbol and forces every $\delta_j = 1$, since for each j another index i exists. □

For $n = 2$, the four words are precisely those printed before Theorem 7.5. The specified ambient group acts on their four types as the square group of order eight. But an arbitrary three-cycle of the four free generators also acts by a correlation, and such a permutation is not in that square group. The full quotient in (F10) is S_4 of order twenty-four. Thus the quotient assertion in Theorem 7.5 is false; it is not repaired merely by giving a more detailed proof of the action of the chosen subgroup.

The geometric conclusions and the chosen finite actions of Theorems 7.5–7.6 are recovered. Their weak-representation conclusions also hold: the displayed subgroup is a finite-rank free group, so Corollary UR.WeakFree embeds its full automorphism group into the type-preserving group, compatibly with its inner subgroup. This supplies a different inner action from the source's chamber-regular construction. The source's argument from the chosen finite signed action does not itself establish the weak conclusion, and the natural inner embedding cannot extend by Corollary R2.NaturalObstruction. Both distinctions are essential; the literal existence conclusion is not refuted by the natural obstruction.

8 Thinness does not imply splitting without chamber connectivity

The last observation in Section 8 of the source is stated for thin, flag-transitive geometries. Its definition of a geometry requires every flag to extend to a chamber, but does not require the chamber graph to be connected. The distinction is necessary.

We use one completely specified finite bipartite graph B. Its vertex set is $\{1, \dots, 896\}$; the complete adjacency lists, together with explicit permutation generators, are supplied in the mandatory file `graph.json`. No graph catalogue or search is a premise. The complete certificate program checks the following finite facts:

1. B is simple, connected and cubic, with parts of size 448 and 1344 edges.

2. The displayed permutations preserve adjacency and generate a subgroup H of order 5376.

3. Starting with vertex 1 and its three neighbors in four separate singleton colors, and all other vertices in a fifth color, repeated neighbor-color refinement has numbers of cells

$$5, 8, 11, 14, 17, 20, 26, 44, 135, 398, 692, 842, 890, 896.$$

4. The subgroup H^+ preserving the two parts has order 2688 and one orbit on the 1344 unordered edges.

5. The 2688 elements exchanging the parts have orders

order	4	12	28
number	1024	896	768.

These are exhaustive finite calculations on the supplied graph and permutations, rather than tests of a sample or of a proposed graph family. Here is why they prove fullness without relying on a graph-automorphism package. At each refinement step a vertex receives its previous color and the multiset of colors of its neighbors. Every automorphism fixing the four initially distinguished vertices fixes every resulting color. The final partition is discrete, so the pointwise stabilizer of that star is trivial. A vertex stabilizer therefore embeds in the symmetric group on its three neighbors and has order at most 6. Consequently

$$|\operatorname{Aut}(B)| \le 896 \cdot 6 = 5376.$$

The explicitly generated subgroup attains this bound, so $H = \operatorname{Aut}(B)$. The final two calculations concern the full automorphism group.

Theorem 5 (A finite thin flag-transitive nonsplit example)**.** *There is a finite rank-three thin, flag-transitive geometry Δ with* 4480 *objects and* 10752 *chambers whose full correlation group does not split over its type-preserving subgroup. More precisely, for $N = C_2^{2240}$ and the full groups just certified,*

$$C(\Delta) = N \rtimes H, \qquad K(\Delta) = N \rtimes H^+, \qquad C(\Delta)/K(\Delta) \cong C_2.$$

The chamber graph has exactly 1344 *connected components.*

Proof. Apply the thin-doubling proposition of the common paper to B. Its type sizes are 896, 896, 2688, so the edge type is recognized. There are two objects over each of the $896 + 1344 = 2240$ underlying vertices and edges, and eight chambers over each edge. The proposition therefore gives all the stated sizes and the exact full group pair. A complement to $K(\Delta)$ would contain an involution in the exchanging coset. Its projection to H would be an exchanging involution, contradicted by the exhaustive order table. This proves nonsplitting. The connected-case splitting theorem of the common paper is unaffected: each of the 1344 edge cubes is one chamber component. □

This refutes the unqualified geometric splitting observation, under the source's stated definition. It is not a classification of groups admitting thin exact representations, nor a solution of the broader geometric representation questions in Section 8. The graph certificate is a required finite premise of this example. The uniform doubling and connected-case arguments are written proofs; no finite sample substitutes for them.

9 Exact scope, attribution and remaining source assertions

The original constructions, their motivating representation problem and their proposed further investigations belong to Leemans, Stokes and Tranchida. The finite normal-pair existence result used here belongs to Díaz Ramos, Molinier and Viruel and is proved in full in common Theorem UR.FinitePairs. No source correction is asserted as a new historical result or as a newly solved question.

The finite examples and free-basis proofs here rework the explicit phase197 manuscripts. The shadow theorem and its four-element projective instance reproduce the phase196 argument with its separately recorded independent review. They are not claimed as mathematics first proved in this edition. The unrestricted panel-permutation construction additionally recovers the weak free-group representation conclusions using a different compatible inner action. This is not inferred from the finite signed subgroup, and no historical priority is claimed. The common recognition theorem was already used in these earlier arguments; the present edition replaces its archived reference by the complete common paper and retains all F010 application results. The shared finite-existence argument is entirely in the common paper; this application has no sibling-paper proof premise. There is no unpaid premise in an old proof bank.

The classical automorphism facts for S_n away from degree six and for A_5 are explicit retained inputs. The positive finite projective repair also retains the fundamental theorem for the full subspace geometry and the standard automorphism-group identification for $\mathrm{PGL}(n, q)$, $n \geq 3$. The new ordered attachment and its full rigidity proof are supplied here; these classical facts do not prove the attachment properties. The exceptional degree-six argument is given here directly. The cited Atlas constructions for A_6, A_7 are retained historical examples, not reconstructions of their individual incidence tables. Their existence also follows from the complete finite normal-pair theorem UR.FinitePairs in the common paper, without identifying its objects with those Atlas models. The source's proposed orthogonal and Suzuki variants are not proved constructions there and remain prospective investigations here.

Source Proposition 7.1 is the elementary coset incidence construction: right multiplication preserves intersection, and intersecting cosets have a common representative, proving transitivity on vertices and incident pairs. The product-intersection and residual-generation criteria quoted as Theorems 7.2 and 7.4 are retained classical results. For every use in the free-basis family, the common paper gives direct tree and residue proofs instead. The interpretation of "independent" in Theorem 7.3 is a freely independent family, as required by its reduced-word argument; the specific conjugate families satisfy this by the explicit immersed finite core. No weaker irredundancy assumption is substituted.

The general Artin–Tits statement is a pointer to forthcoming work in the source and is not a premise of these free-basis proofs. The background Outer Space realization and maximum finite-subgroup discussion are not needed to construct the signed permutation subgroup. In particular, torsion-freeness of the inner subgroup by itself does not say that every finite outer subgroup lifts to an isomorphic automorphism subgroup. The finite signed actions proved here make no such inference. The source's infinite-type and more restrictive finite-geometry questions remain questions. The common connected-chamber splitting proposition supplies a valid sufficient hypothesis. The complete thin-doubling construction and finite certificate refute the unqualified splitting observation; they do not resolve the remaining representation questions.

The inherited finite programs calculate all groups and flag sets in their stated bounded models; the free-word program checks 45,846 reduced words. The uniform proofs above do not infer infinite statements from those samples. The projective program independently constructs all 2562 objects and their fibers, but the written counting argument proves the displayed obstruction without relying on the program's output. The full files are retained as optional validation. The three basis-permutation Lean certificates prove the action on free generators, transport of each complementary generated subgroup and faithfulness. The seven inherited shadow declarations prove the actual pointwise incidence kernel and its triviality criterion; they do not prove the image or splitting of restriction. Neither file formalizes the projective instance, free-basis geometry, residual connectedness or normalizer. The twelve previously accepted F010 certificates remain scoped as in the first application.

References

[1] D. Leemans, K. Stokes and P. Tranchida, *Groups represented by incidence geometries*, arXiv:2506.10464v1, 12 June 2025.

[2] A. Díaz Ramos, R. Molinier and A. Viruel, *Every finite group is represented by a finite incidence geometry*, arXiv:2511.02410v2, 16 December 2025.

[3] F. Buekenhout and A. M. Cohen, *Diagram Geometry*, Springer, 2013, Theorem 1.8.10 and Corollary 1.8.13, as cited in the source.

[4] F. Buekenhout, M. Dehon and D. Leemans, *An atlas of residually weakly primitive geometries for small groups*, Acad. Roy. Belg., 1999; retained examples as cited in the source.